

1. What is a Single Page Application (SPA)?

A **Single Page Application (SPA)** is a web application that loads a single HTML page initially and dynamically updates the content without reloading the entire page.

Instead of requesting a new webpage from the server every time the user clicks a link, JavaScript updates only the required components.

Examples

- Gmail
- Facebook
- Instagram
- Google Maps
- Netflix

Benefits of SPA

- Faster user experience
- No full page reloads
- Reduced server load
- Better performance
- Smooth navigation
- Easy API integration
- Better user interaction

2. What is React?

React is an **open-source JavaScript library** developed by **Meta (Facebook)** for building interactive user interfaces.

It is mainly used to develop:

- Single Page Applications
- Dynamic websites

- Mobile applications (React Native)

React is component-based, meaning applications are built using reusable UI components.

How React Works

React follows these steps:

1. User performs an action.
2. State changes.
3. React creates a Virtual DOM.
4. Virtual DOM compares with previous Virtual DOM.
5. Only changed elements are updated in the Real DOM.

This process is called **Reconciliation**.

SPA vs MPA

Feature	SPA	MPA
Page Reload	No	Yes
Speed	Faster	Slower
User Experience	Better	Moderate
Initial Load	Slightly Heavy	Light
Server Requests	Less	More
SEO	More Difficult	Easier
Examples	Gmail, Facebook	Amazon, Flipkart (traditional pages), Banking portals

Pros of SPA

- Fast navigation
- Better performance
- Less bandwidth usage

- Rich user experience
- Easy REST API integration
- Reusable components

Cons of SPA

- Initial loading may take longer
- SEO is more challenging
- JavaScript dependency
- Browser memory usage can increase

What is Virtual DOM?

The **Virtual DOM** is a lightweight JavaScript copy of the Real DOM.

Instead of updating the actual webpage directly, React:

1. Creates a Virtual DOM.
2. Compares old and new Virtual DOM.
3. Finds only the changed elements.
4. Updates only those elements.

This makes React applications much faster.

Features of React

- Component-Based Architecture
- JSX Syntax
- Virtual DOM
- One-Way Data Binding
- High Performance
- Reusable Components
- Easy Learning Curve
- Strong Community Support

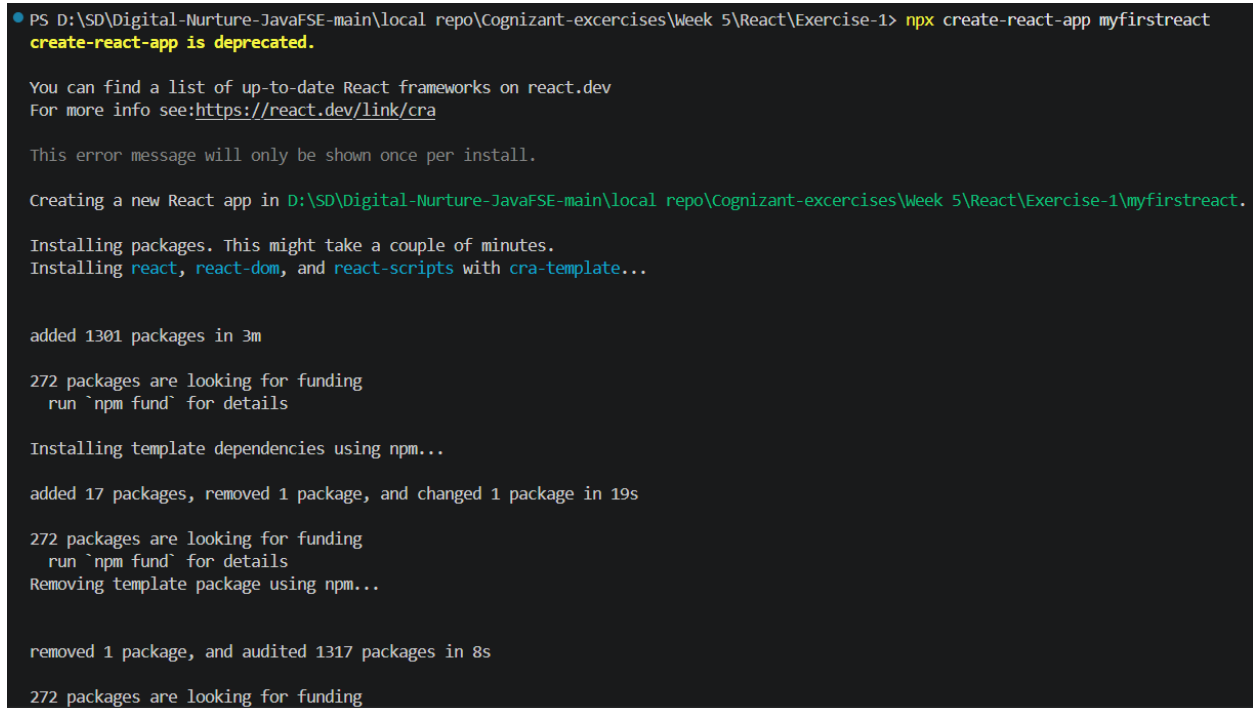
- Cross-platform development using React Native

Practical:

App.js:

```
function App() {  
  
  return (  
  
    <div>  
  
      <h1>Welcome to the first session of React</h1>  
  
    </div>  
  
  );  
}  
  
export default App;
```

Output Screenshots:



```
PS D:\SD\Digital-Nurture-JavaFSE-main\local repo\Cognizant-exercices\Week 5\React\Exercise-1> npx create-react-app myfirstreact  
create-react-app is deprecated.  
  
You can find a list of up-to-date React frameworks on react.dev  
For more info see:https://react.dev/link/cra  
  
This error message will only be shown once per install.  
  
Creating a new React app in D:\SD\Digital-Nurture-JavaFSE-main\local repo\Cognizant-exercices\Week 5\React\Exercise-1\myfirstreact.  
  
Installing packages. This might take a couple of minutes.  
Installing react, react-dom, and react-scripts with cra-template...  
  
added 1301 packages in 3m  
  
272 packages are looking for funding  
  run `npm fund` for details  
  
Installing template dependencies using npm...  
  
added 17 packages, removed 1 package, and changed 1 package in 19s  
  
272 packages are looking for funding  
  run `npm fund` for details  
Removing template package using npm...  
  
removed 1 package, and audited 1317 packages in 8s  
  
272 packages are looking for funding
```

```
We suggest that you begin by typing:

  cd myfirstreact
  npm start

Happy hacking!
PS D:\SD\Digital-Nurture-JavaFSE-main\local repo\Cognizant-exercices\Week 5\React\Exercise-1> cd myfirstreact
PS D:\SD\Digital-Nurture-JavaFSE-main\local repo\Cognizant-exercices\Week 5\React\Exercise-1\myfirstreact> npm start

> myfirstreact@0.1.0 start
> react-scripts start

(node:7508) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAfterSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
(Use 'node --trace-deprecation ...' to show where the warning was created)
(node:7508) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onBeforeSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
Starting the development server...
Compiled successfully!

You can now view myfirstreact in the browser.

  Local:            http://localhost:3000
  On Your Network:  http://192.168.29.126:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
```

